

SparkADS

SparkADS is a Beckhoff TwinCAT 2/3 ADS/AMS driver that talks ADS directly to the PLC. No TF6100 OPC UA server license is required.

Version 0.1.1 Ignition 8.1.33+ and 8.3 Runs on: any gateway that reaches the PLC network

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1. Overview

SparkADS is a standalone device driver, licensed and running independently of any other SparkBridge module. Module id `com.greenguardusa.sparkads`.

2. What You Need

- Ignition 8.1.33 or newer, or the 8.3 line (separate build).
- A TwinCAT 2 or 3 PLC reachable over TCP 48898 (ADS/AMS), with a static AMS route configured on the PLC pointing to the gateway's AmsNetId.

3. Install

Install the signed module on Ignition 8.1 or 8.3 through Config > Modules > Install or Upgrade a Module; no build step is required. Module id `com.greenguardusa.sparkads`.

4. License

Licensed per gateway, bought once and kept, soft-licensed (`com.greenguardusa.license`, entitlement ADS).

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at `<data>/sparkbridge-license.key`; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

5. Configure

Config > OPC UA > Device Connections > Create new Device > the ADS device type.

Setting	Meaning, default
Hostname/IP, Port	PLC address; AMS/TCP port 48898.
Target NetId, Target Port	The PLC's AmsNetId and AMS port. TwinCAT 3 PLC1 defaults to port 851 (801/811/821/831 are TwinCAT 2 runtimes 1-4; 851/852/853 are TwinCAT 3 runtimes 1-3).
Local NetId	This driver's own AmsNetId, which must be added as a route on the PLC.

Manual tag addressing: `MAIN.x` (symbolic), `%MW100:INT`, or `0x4020,0x100:DINT` (index group/offset).

6. Operate

The PLC must have a static AMS route pointing to the gateway's Local NetId before the device can connect; a device that never leaves DISCONNECTED usually means the route is missing or the Target NetId/Port do not match the PLC's runtime.

7. Verify

See the README in the bundle for the verification procedure.

8. Troubleshoot

Symptom	Cause / fix
Device will not connect	Confirm the AMS route on the PLC points to the gateway's Local NetId, and that Target NetId/Target Port match the PLC's actual runtime (TwinCAT 2 vs 3, PLC instance number).
Symbolic address not found	Confirm the symbol name matches the PLC project exactly, including case, for example <code>MAIN.x</code> .

9. Reference

Module id	<code>com.greenguardusa.sparkads</code>
Default port	TCP 48898 (ADS/AMS)
TwinCAT 3 PLC1 default port	851
License entitlement	ADS (<code>com.greenguardusa.license</code>)

10. Support

Docs page section: </sparkbridge/docs#sparkads>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).